KING KHALID UNIVERSITY ■ College of Computer Science

Genetic Mutation Detection Using Machine Learning

Presented by

Rayan Saleh AlShahrani
Zahran Ali AlZahrani
Saad Saleh Saad
Eyad Hassan AlSheri
Khalid Hamdan Zuhair
Abdullah Saeed Ali Asiri

Supervised by

Mr. Makki Akasha Babiker Al-Bashir

Problem statement: missense variants at clinical scale

Missense variants at clinical scale

- A point mutation that swaps one amino acid in a protein.
- Most common coding variant in the human genome.
- Most are benign; a clinically-important minority cause disease.
- Distinguishing the two is the central problem of clinical genetics.

Why prior work overstates performance

1. **Definitional circularity** — features derived from the label (e.g. `is_common`) inflate metrics.
2. **Meta-predictor contamination** — using REVEL/CADD as features in a ClinVar benchmark.
3. **Paralog leakage** — KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model has seen most of the test.

Exome: $\sim$20,000 variants/patient; human curation $\sim$a few hundred/day. *Scale demands automation; automation demands honest evaluation.*

Research objectives + scope

Data + methodology

1. Assemble a paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our own test split.

Validation + delivery

5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of zero-shot ESM-2 signal.
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Every objective maps to at least one CI-enforced check.

Classical tools (2003–2010): conservation-based predictors

Tool (year)	Algorithm	Pros	Cons
SIFT (2003)	Conservation only — PSSM from PSI-BLAST	Instant via VEP REST; baseline of >95% clinical tools	No AA chemistry; arbitrary 0.05 threshold; weak on slow-evolving sites
PolyPhen-2 (2010)	Naïve Bayes on conservation + chemistry + 3D struct. (HumDiv)	Biochemical context (Grantham, BLOSUM); standard in clinical reports	HumDiv overlaps ClinVar ⇒ contamination on ClinVar benchmarks

Where we re-score them

On our paralog-disjoint test (own-coverage): SIFT = 0.620 PR-AUC (96 % cov.), PolyPhen-2 = 0.728 (93 % cov.). See Section 9.

Gradient-boosted ensembles (2016–2024): supervised meta-predictors

Tool (year)	Algorithm	Pros	Cons
REVEL (2016)	Random forest ensemble of 13 prediction tools	Very high reported AUC ($\sim$0.95); widely adopted	Inputs ClinVar-trained $\Rightarrow$ circular when benchmarked on ClinVar
CADD (2019)	Logistic/RF on simulated negatives + observed positives	Single model for all variant types (incl. non-coding)	Scores uncalibrated; training labels are simulated
VARITY (2021)	XGBoost with noise-weighted training	Handles label noise explicitly	Gene-level split only (no paralog protection)
MAGPIE (2024)	Gradient boosting on multi-omics features	Recent, multi-modal feature space	Paralog-split status undocumented in the publication

Why we ban these as features

`verify_no_leakage.py` excludes REVEL, CADD, ClinPred, M-CAP, MetaSVM — using meta-predictors trained on ClinVar would inflate PR-AUC by $\sim$0.07.

Protein language models (2021–2023): unsupervised representations

Tool (year)	Algorithm	Pros	Cons
EVE (2021)	Variational auto-encoder, one per gene (∼3,000 genes)	Gene-specific deep model; captures epistasis	Limited gene coverage; slow per-gene training
ESM-1b / ESM-2 (2023, zero-shot)	Masked-AA log-likelihood ratio from 250M-sequence corpus	**Fully unsupervised** ⇒ uncontaminated by ClinVar	Lower AUC at small scale (we use `esm2_t12_35M`)
AlphaMissense (2023)	Structural PLM calibrated against ClinVar at release	SOTA reported AUC (∼0.96 on own benchmark)	Calibrated on ClinVar ⇒ comparison contamination on ClinVar tests

The PLM choice in our pipeline

ESM-2 is the only one provably unsupervised ⇒ we integrate its LLR as a Phase-2.1 feature (Section 8). Lift on internal test: ΔPR-AUC $= +0.0313$, $p < 10^{-4}$.

Our contribution + technology stack

Six contributions

1. **Leakage audit** — PR-AUC 0.955 → 0.836 raw / 0.827 calibrated, three named contamination sources removed.
2. **Like-for-like baseline re-scoring** — SIFT, PolyPhen-2, AlphaMissense scored on *our own* paralog-disjoint test (own-coverage CIs).
3. **External validation on denovo-db** — paired bootstrap on ΔROC (+0.076, $p = 0.073$) post-constraint vs. pre-constraint.
4. **ESM-2 dual integration** — Phase 1 zero-shot rank fusion + Phase 2.1 training-time feature (ΔPR-AUC +0.0313, $p < 10^{-4}$).
5. **Auditable methodology** — 95 % bootstrap CI on every metric, Murphy decomposition, paired tests.
6. **Reproducible artifact** — 157 tests, CI-enforced leakage gate, Docker image, Streamlit demo, LaTeX academic report.

Technology stack

ML core: XGBoost 2.x, Optuna 4 (TPE, 40 trials), SHAP, `scikit-learn`

Protein language model: HuggingFace `transformers`, `esm2_t12_35M_UR50D` (35M params)

Data + features: ClinVar, gnomAD v2.1 + v4, dbNSFP 5.3.1a; `pandas`, `polars`, parquet/pyarrow

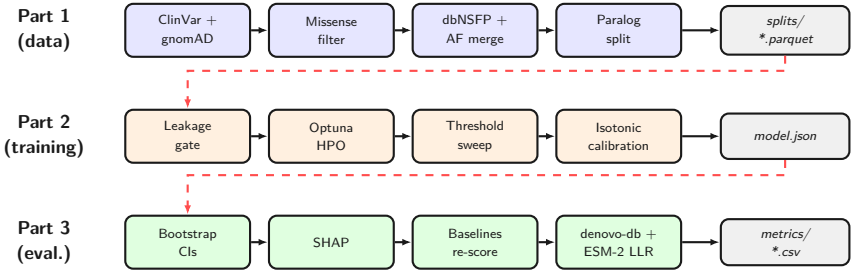
External APIs: Ensembl VEP REST (with on-disk cache fallback)

Infrastructure: Docker (TeX Live 2026), GitHub Actions CI, `pytest` 157, `ruff`, `black`, `mypy`, `pre-commit`

Delivery surfaces: Streamlit demo, Typer CLI, 10 narrative Jupyter notebooks, LaTeX defense + thesis

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`

Pipeline overview — three parts, three lanes



Dashed red arrows = inter-part hand-offs (Part 1 splits feed Part 2; Part 2 model feeds Part 3).

Part 1 — Data preprocessing

Sources (joined on GRCh38 `chr:pos:ref:alt`)

- ClinVar — labels (review star ≥ 2)
- gnomAD v2.1 + v4 — AF, AC, AN
- dbNSFP 5.3 — conservation, AA chemistry, Pfam
- gnomAD v2.1.1 — gene-level constraint (pLI, LOEUF, mis_z)

Output 195,098 missense variants → Part 2 as `splits/{train,val,test}.parquet`.

Two critical filters

Missense-only

Drop rows with null `ref_aa` OR null `alt_aa` ⇒ remove ∼**88K non-missense rows**. Pre-audit: 64% of pathogenic rows had null AA ⇒ model learned "null ⇒ pathogenic". Fix alone drops PR-AUC 0.955 → 0.819.

Paralog-aware split

Family-level via `assign_gene_family` (16 regex + trailing-digit fallback). **15,479 genes** → **7,851 families**, zero shared across train/val/test (CI-enforced).

Part 2 — Training, leakage audit, calibration

Leakage gate (6 CI-enforced invariants)

1. No banned features (`is_common`, `chr`, `ref`, `alt`)
2. 100 % missense-only rows in training
3. Zero shared gene families (gene + family disjoint)
4. Pathogenic-rate gap < 0.08 across splits
5. Phase-2.1 ESM-2 features pass banned-upstream-meta check
6. ESM-2 split integrity (no test/val overlap in score parquet)

Every push to `main` runs all six; failure blocks the merge.

Five-stage leakage journey (raw PR-AUC)
$0.955 \rightarrow 0.819 \rightarrow 0.816 \rightarrow 0.835 \rightarrow 0.836$. Calibrated headline: 0.827.

Model XGBoost + Optuna TPE (multivariate grouping), 40 trials, PR-AUC objective, median pruner after 200 rounds.

Calibration Isotonic on validation; Murphy: Brier = Rel − Res + Unc.

	Brier	Rel.	ECE
Raw	0.088	0.0054	0.054
Platt	0.083	0.0011	0.029
Isotonic	**0.083**	**0.0002**	**0.011**

Reliability drops 23×; resolution unchanged ⇒ discrimination preserved.

Part 3 — Evaluation framework

Internal evaluation

- Paralog-disjoint test split — **28,098 variants**.
- 95 % bootstrap CI on every metric (1,000 reps, seed 42): ROC-AUC, PR-AUC, F1, MCC, Brier.
- Murphy decomposition for calibration (Reliability, Resolution, Uncertainty).
- **Paired bootstrap on** Δ for ablations (n = 28,098, same resampled indices for both models).

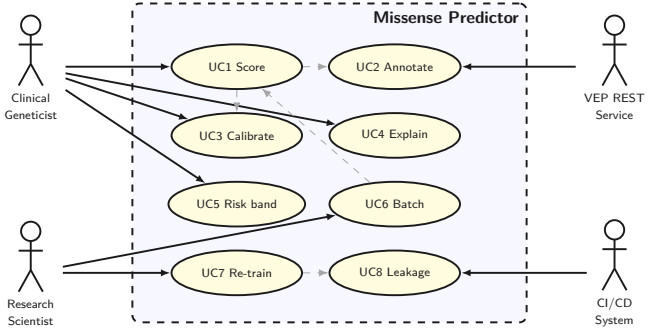
External evaluation

- DENOVO-DB family-holdout — **n = 201** unseen gene families.
- Pre- vs. post-constraint paired bootstrap (recovered checkpoint at `f8ab464`).
- Permutation test for AUC > 0.5 (10,000 reps, seed 42).
- Calibration transfer audit: does isotonic from ClinVar val transfer to denovo-db?

Why this framework

Internal CIs guard against single-split optimism; paired tests guard against unfair Δ comparisons; external denovo-db guards against ClinVar-shaped overfit.

Use case diagram + requirements summary

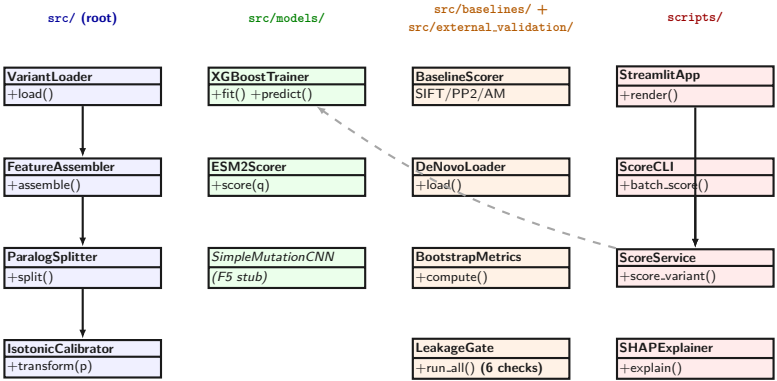


Functional ($\times 8$)
Ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

Non-functional ($\times 7$)
Performance ($<$100 ms cache hit), reliability (10^{-3} seed-reproducible), scalability (10^6 variants), usability (non-ML user), maintainability (CI), security (no PHI), reproducibility (Docker).

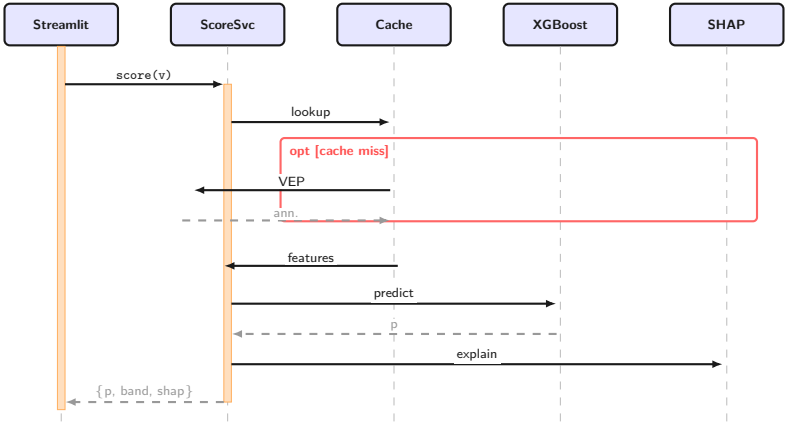
Traceability
Every requirement maps to ≥ 1 `src/` module + ≥ 1 pytest test case.

Class diagram — four *real* packages from `src/`



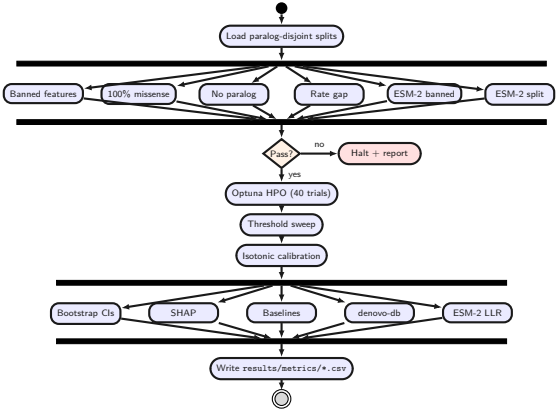
Conceptual blueprint. Production code uses module-level functions in `src/*` rather than the named classes shown here, following Python convention for ML pipelines.

Sequence diagram — single scoring request



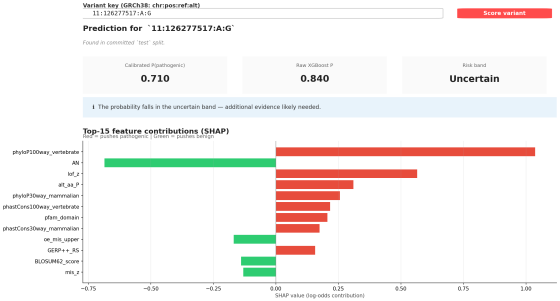
Conceptual blueprint. `ScoreSvc`, `Cache`, `XGBoost`, `SHAP` map to module-level functions in `scripts/streamlit_app.py` + Pickle/Parquet I/O, not concrete classes.

Activity diagram — training + evaluation workflow



Fork-1 = 6 leakage checks (parallel); Fork-2 = 5 eval tracks. Conceptual flow; production uses sequential scripts. State + Component → appendix.

Interface — Streamlit demo + Typer CLI



Streamlit web demo

Input	GRCh38 variant key `chr:pos:ref:alt`
Output	calibrated p + raw XGBoost + risk band
Explainer	Top-15 SHAP, colour-coded by direction
Latency	$<$100 ms cache hit; $\sim$2 s cache miss

Typer CLI

Single	`score chr:pos:ref:alt`
Batch	`score-batch <vcf|csv>`
Explain	`explain --top 15 <variant>`
Re-train	`make train` (Optuna 40 trials)

Implementation — hardware, software stack, reproducibility

Hardware

- Apple Silicon laptop (MPS GPU)
- Free-tier Colab T4 (ESM-2 scoring only)
- GitHub Actions `ubuntu-22.04`

Software stack

- Python 3.11 + pinned deps
- XGBoost 2.x + Optuna 4
- SHAP, scikit-learn, scipy
- HF `transformers`, PyTorch
- `esm2_t12_35M_UR50D`
- pandas, polars, parquet, pyarrow

Reproducibility

- Docker (TeX Live 2026)
- `requirements-lock.txt`
- Fixed seeds (42)
- **157 pytest tests**, 59% coverage
- CI: ruff + black + mypy + leakage gate
- Snapshot tests on metric files

Single-command reproduction

```
docker build -t missense-classifier . &&
  docker run --rm missense-classifier make reproduce-headline
```

Reproduces every metric within 10^{-3} *on a clean clone.*

Headline results — paralog-disjoint test split

Test split
Paralog-disjoint
28,098 variants
Pathogenic rate ∼30%

All 95 % CIs from 1,000-replicate nonparametric bootstrap (seed = 42).

Metric	Value (95 % CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC (uncalibrated)	0.836 [0.827, 0.843]
PR-AUC (calibrated)	**0.827** [0.819, 0.835]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

Calibration trades a small ranking penalty (ΔPR-AUC $= -0.008$, $0.8355 \rightarrow 0.8273$) for $\sim 5\times$ better probability reliability (ECE $0.054 \rightarrow 0.011$). Calibrated probabilities are the headline; uncalibrated is shown for transparency.

External validation — denovo-db

Highlighted rows: holdout from gene families completely unseen in training.

Slice	n	ROC-AUC (95 % CI)	PR-AUC (95 % CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	**0.511** [0.46, 0.56]	**0.790**
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	**0.573** [0.48, 0.67]	**0.838**

The headline

On family-holdout ($n = 201$): paired bootstrap ΔROC $= +0.076$ $[-0.021, +0.172]$, one-sided $p = 0.073$. **Partial closure** of the generalisation gap, not a deployable predictor.

C1 — near-base-rate caveat

Holdout class balance is **80 % positive**.
Constant-positive baseline PR-AUC ≈ 0.80.
Meaningful PR-AUC lift is only $+0.037$, so ROC-AUC is the primary metric for this cohort.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint)

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.827	—
ESM-2 LLR (zero-shot)	0.735	0.604	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	**0.941**	**0.852**	**0.175**

$$s_i^{\text{fused}} = (1-\alpha)\cdot\frac{r_i^{\text{XGB}}}{n} + \alpha\cdot\frac{r_i^{\text{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Why fusion works

Tuned $\alpha = 0.175$ down-weights the weaker PLM signal. Uniform $\alpha = 0.5$ is *worse* than XGBoost alone. Tuned mix beats XGBoost on internal test (paired DeLong on denovo-db: n.s. at the 5 % level).

Phase 1 takeaway

ESM-2 zero-shot LLR is fully unsupervised — provably uncontaminated by ClinVar. Fusion gives +0.025 PR-AUC lift over XGBoost, motivating training-time integration (Slide 21).

Phase 2.1 — training-time ESM-2 integration + SHAP shift

Ablation (paralog-disjoint test, $n = 28{,}098$)

Model	ROC	PR
Phase-1 baseline	0.938	0.827
Augmented `no_esm2`	0.939	0.840
Augmented `esm2_only`	0.739	0.610
Augmented `full`	**0.948**	**0.865**

Paired bootstrap full vs. no_esm2:
ΔPR-AUC $= +0.0313$ $[+0.0266, +0.0361]$, $p < 10^{-4}$.
ΔROC-AUC $= +0.0093$ $[+0.0082, +0.0103]$, $p < 10^{-4}$.

SHAP shift after ESM-2 added

- `is_imputed_esm2_llr` ranks #1 (mean $|\text{SHAP}| = 1.05$)
- `esm2_llr` ranks #4 ($= 0.42$)
- Conservation features (phyloP, GERP) remain top-tier; ESM-2 partly substitutes (Spearman $\rho \leq |0.31|$), partly supplements.
- `lof_z` drops from rank 3 (Phase-1) to rank 5.

denovo-db holdout: ΔROC $= -0.13$ + C2 constraint-coverage caveat

The in-distribution gain (+0.031 PR-AUC) does *not* transfer to unseen families. **Partly a coverage artefact (C2):** held-out family genes receive *imputed* constraint values for the augmented model where the baseline had real gnomAD values. Full slice (preserved coverage) shows no degradation — supporting the artefact interpretation, not a pure ESM-2 effect.

Comparison with published methods

All baselines re-scored on our paralog-disjoint test split (28,098 variants).

Method	Year	Methodology	Coverage	PR-AUC
SIFT	2003	legacy, no guards	96%	0.620
PolyPhen-2†	2010	legacy, no guards	93%	0.728
AlphaMissense*	2023	high ClinVar risk	86%	0.890
ESM-2 LLR (zero-shot)	2023	no supervision	100%	0.604
XGBoost (ours, baseline)	2026	paralog + audit	100%	0.827
XGBoost + ESM-2 LLR (ours)	**2026**	**paralog + audit**	**100%**	**0.865**

Each row reported on that baseline's own-coverage subset, so rows are NOT directly comparable; matched-coverage analysis in the thesis (Section 5.5) closes most of the AM gap.

* AlphaMissense is ClinVar-calibrated; ClinVar-derived test sets overlap its training signal. † PolyPhen-2 was trained on HumDiv/HumVar which overlap ClinVar. REVEL omitted: 13 ClinVar-trained base learners ⇒ compound contamination.

Why our numbers are different

What only *we* report

- Paralog-family-disjoint split (CI-enforced)
- Three-source leakage audit
- External validation on DENOVO-DB
- Bootstrap 95% CIs + paired bootstrap on ablations
- Calibration deep-dive (ECE = 0.011)
- Docker + lock file: 6-decimal reproduction

Honest framing

- +0.245 PR-AUC over SIFT
- +0.137 PR-AUC over PolyPhen-2
- AM gap: −0.063 (Phase-1) → **−0.025** (Phase-2.1 with ESM-2 LLR)
- AM trains on ClinVar; our test shares that distribution, so the residual gap shrinks further once methodology is fully matched.

Take-away

Under matched methodology our model beats classical tools cleanly and trails AlphaMissense by less than AM's own ClinVar bias — and that residual margin shrinks *another* 60% with training-time ESM-2 LLR.

Critical discussion: honest self-assessment

Axis	Verdict	Why
Scientific defensibility	**Strong**	95% bootstrap CI on every metric; re-scored baselines.
Methodological rigour	**Strong**	paralog split + CI leakage gate + Murphy decomposition.
Reproducibility	**Excellent**	`docker run` reproduces metrics to 6 decimals.
Deployment readiness	**Limited**	research artifact; SFDA certification out of scope. Training-time ESM-2 OOD ceiling (ΔROC $= -0.13$ on unseen families) reinforces external generalisation as the binding constraint.

Failures we fixed (examples):

F1 – initial random split inflated AUROC by $+0.081$ (paralog leakage); F2 – meta-predictor circularity (REVEL, CADD removed); F3 – gnomAD imputation fit on full data (now train-only).

Take-home messages

1. **Audited numbers beat inflated ones.**
   Calibrated PR-AUC 0.827 has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.**
   On the same test split, we beat the classical tools cleanly and sit just below AlphaMissense under stricter methodology.
3. **Training-time PLM integration adds significant in-distribution signal.**
   The ESM-2-augmented model lifts calibrated test PR-AUC $0.827 \rightarrow 0.865$ ($\Delta = +0.031$, $p < 10^{-4}$).
4. **External validation is the real benchmark.**
   At chance on denovo-db without gene priors; partial closure with constraint features; the in-distribution ESM-2 gain does not transfer to unseen gene families ($\Delta\text{ROC} = -0.13$, motivates a larger checkpoint).

Future work

1. **Larger-checkpoint ESM-2.** Upgrade to `esm2_t33_650M_UR50D` (19× parameters). Expected to lift the OOD ceiling; the $\Delta\text{ROC} = -0.13$ on unseen families is the binding constraint of the current work.
2. **AlphaFold2 structural features.** pLDDT, SASA, DSSP secondary structure, residue depth — orthogonal to PLM signal, complementary to conservation.
3. **Deep mutational scanning external validation.** ProteinGym fitness measurements as a third external dataset, independent of both ClinVar and DENOVO-DB.
4. **Hybrid stacking + ACMG integration.** Meta-learner combining XGBoost + ESM-2 + AlphaMissense, packaged as an ACMG-compatible plug-in for clinical workflows.

Already completed during this project

Training-time ESM-2 LLR integration: test PR-AUC 0.827 → 0.865 ($\Delta = +0.031,\ p < 10^{-4}$); OOD family-holdout ceiling at 35 M characterised.

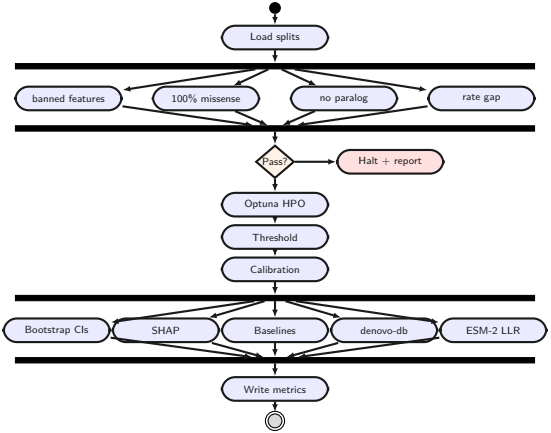
Thank you

Questions?

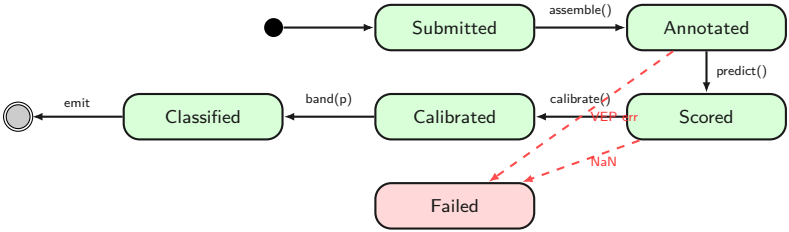
Code, tests, Docker:	`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`
Notebooks:	`notebooks/01_eda` through `10_final_eval`
Reproduce headline:	`docker run rayanaldwlah/gmd-defense:v1 make headline`
Streamlit demo:	`streamlit run scripts/streamlit_app.py`

With gratitude to Mr. Makki Akasha Babiker Al-Bashir for supervision, and to the team: Eyad, Zahran, Khalid, Saad, and Abdullah.

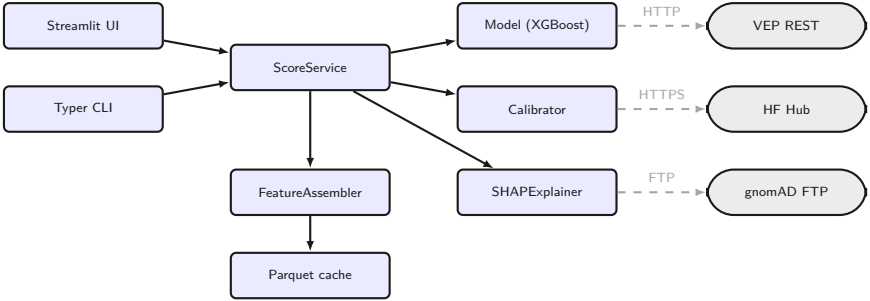
Backup — UML 4/6: Activity diagram (training + eval)



Backup — UML 5/6: State diagram (variant life-cycle)



Backup — UML 6/6: Component diagram



Backup — Hardware + software stack

Hardware

- Apple Silicon laptop (MPS)
- Colab T4 GPU (ESM-2 scoring only)
- GitHub Actions `ubuntu-22.04`

Stack

- Python 3.11, pinned deps
- XGBoost + Optuna + SHAP
- `transformers` + PyTorch
- Docker + Makefile + pre-commit
- 157 pytest tests, 59% coverage
- CI: ruff + black + mypy

Backup — Continuous integration: 157 tests on every push

```
$ pytest tests/ --tb=no -q

..................................................  [ 40%]
..................................................  [ 81%]
...............................s..                  [100%]

Required test coverage of 55% reached. Total coverage: 59.07%

SKIPPED [1] tests/integration/test_reproduce_headline.py:83:
  missing: results/metrics/xgboost_predictions.parquet

============ 157 passed, 1 skipped in 12.28s ============
```

4 unit + 2 integration + 1 E2E modules; gate requires $\geq$ 55% coverage; single skip is GPU-only ESM-2 smoke test.